

02624 Inverse Problems and Imaging

Generalized Tikhonov and Total Variation

Yiqiu Dong

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Regularization in Finite Dimension

Consider to solve ill-posed problem:

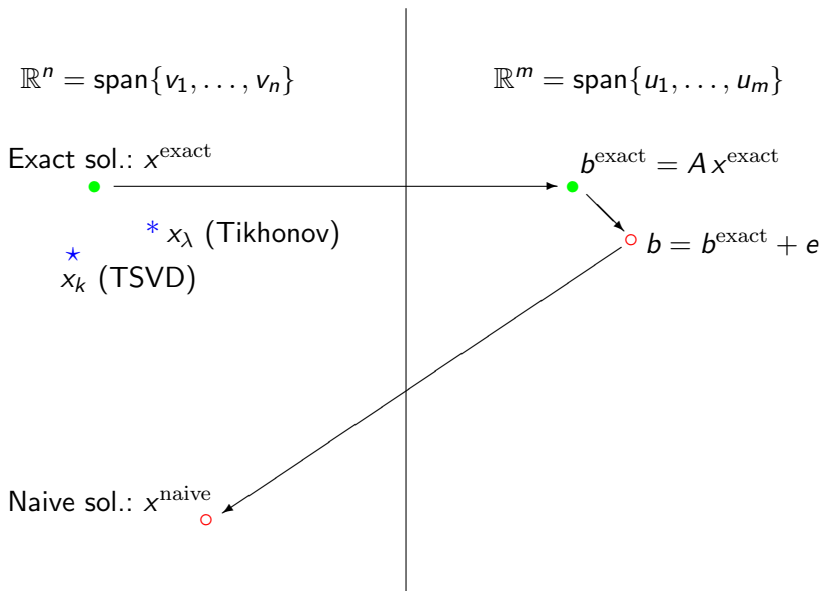
$$Ax = b = b^{\text{exact}} + e.$$

Regularization: Approximate the inverse matrix, A^{-1} , by a set of well-conditioned matrices $\mathcal{R}_\lambda$, where λ is the regularization parameter.

We need:

- With the noise-free data, the matrix $\mathcal{R}_\lambda$ with $\lambda = 0$ should give A^{-1} , and $x_\lambda = \mathcal{R}_\lambda b^{\text{exact}}$ should give x^{exact} .
- With the noisy data, we can find an appropriate parameter λ such that $x_\lambda = \mathcal{R}_\lambda b$ is a good approximation of x^{exact} .

Illustration of the Need for Regularization



Truncated SVD

A simple way to reduce the influence of the noise is to discard the SVD coefficients where the noise starts to be dominant.

Define *truncated SVD* (TSVD) solution as

$$\begin{aligned}x_k &= \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i, & k < n \\&= \sum_{i=1}^n \varphi_i \frac{u_i^T b}{\sigma_i} v_i,\end{aligned}$$

where $\varphi_i = 1$ if $1 \leq i \leq k$, otherwise $\varphi_i = 0$.

Truncated SVD

The solution of Truncated SVD is also the solution of

$$\min \|x\|_2^2, \quad \text{s.t.} \quad \|A_k x - b\|_2^2 = \min$$

with approximating A by the rank- k matrix

$$A_k = U \Sigma_k V^T = \sum_{i=1}^k \sigma_i u_i v_i^T, \quad k < l,$$

where $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0)$.

- **Advantages:**

- Intuitive
- Easy to compute, *if we have the SVD*

- **Drawback:**

- For large-scale problem, it is infeasible to compute the SVD

Idea: If we control the norm of the solution, then we should be able to suppress most of the large noise components.

The Tikhonov solution x_λ is defined as the solution to

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2$$

- $\|Ax - b\|_2^2$: the residual term (the data-fitting / data-fidelity term)
- $\Omega(x) = \|x\|_2^2$: the regularization term
- λ large: strong regularization, over smoothing.
- λ small: good fitting

SVD and Tikhonov Regularization

Two alternative formulations

$$\min_x \left\| \begin{pmatrix} A \\ \lambda I \end{pmatrix} x - \begin{pmatrix} b \\ 0 \end{pmatrix} \right\|_2^2$$

$$(A^T A + \lambda^2 I) x = A^T b$$

We can write the discrete Tikhonov solution x_λ in terms of the SVD of A as

$$x_\lambda = (A^T A + \lambda^2 I)^{-1} A^T b = \sum_{i=1}^n \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2} \frac{u_i^T b}{\sigma_i} v_i = \sum_{i=1}^n \phi_i^{[\lambda]} \frac{u_i^T b}{\sigma_i} v_i.$$

The filter factor $\phi_i^{[\lambda]}$ dampens the components in the solution corresponding to small σ_i .

Tikhonov Regularization in General Form §8.1

The Tikhonov regularization problem is

$$\min_x \{ \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2 \} .$$

It obtains stability by control the norm of the solution.

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We can obtain a general version by replacing the norm $\|x\|_2$ by the form $\|Lx\|_2$, where L is a discrete approximation of a derivative operator.

The Tikhonov regularization problem in *general form* is thus

$$\min_x \{ \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2 \} .$$

The matrix L is $p \times n$ with no restrictions on the dimension p .

If L is invertible, such that L^{-1} exists, then the solution can be written as

$$x_{L,\lambda} = L^{-1}\bar{x}_\lambda$$

where $\bar{x}_\lambda$ solves the standard-form Tikhonov problem

$$\min_{\bar{x}} \{ \|(A L^{-1}) \bar{x} - b\|_2^2 + \lambda^2 \|\bar{x}\|_2^2 \}.$$

The multiplication with L^{-1} in the back-transformation $x_\lambda = L^{-1}\bar{x}_\lambda$ represents integration, which yields additional smoothness in the Tikhonov solution, compared to $L = I$.

About the L Matrix

Similar to the standard-form problem obtained for $L = I$, the general-form Tikhonov solution $x_{L,\lambda}$ is the solution to a linear least-squares problem:

$$\min_x \left\| \begin{pmatrix} A \\ \lambda L \end{pmatrix} x - \begin{pmatrix} b \\ 0 \end{pmatrix} \right\|_2.$$

The solution $x_{L,\lambda}$ is unique when the coefficient matrix has full rank, i.e., when the null spaces of A and L intersect trivially:

$$\mathcal{N}(A) \cap \mathcal{N}(L) = \{0\}.$$

Since multiplication with A represents a smoothing operation, it is unlikely that a null vector of L (if L is rank deficient) is also a null vector of A .

About the L Matrix

Two common choices of L are the rectangular matrices

$$L_1 = \begin{pmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} \in \mathbb{R}^{(n-1) \times n}$$

$$L_2 = \begin{pmatrix} 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \end{pmatrix} \in \mathbb{R}^{(n-2) \times n}$$

which represent the first and second derivative operators.

Thus, the discrete smoothing norm $\|Lx\|_2$, with L given by either I , L_1 or L_2 , represents the continuous smoothing norms $\|x\|_2$, $\|x'\|_2$, and $\|x''\|_2$, respectively.

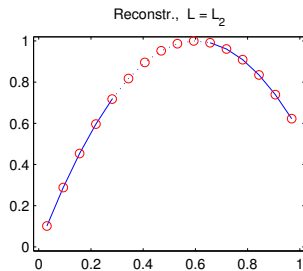
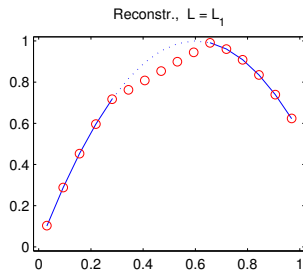
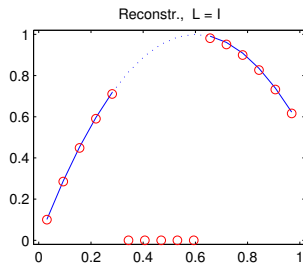
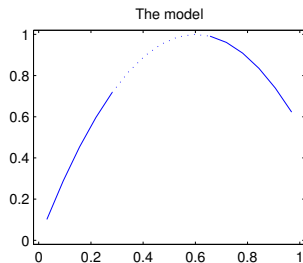
Let x be given as samples of a function, and let the right-hand side be given by a subset of these samples, e.g.,

$$b = Ax, \quad A = \begin{pmatrix} I_{\text{left}} & 0 & 0 \\ 0 & 0 & I_{\text{right}} \end{pmatrix},$$

where I_{left} and I_{right} are two identity matrices.

The figure next page shows with the three discrete smoothing norms $\|x\|_2$, $\|L_1 x\|_2$ and $\|L_2 x\|_2$ how the missing data are filled in.

Example



Tikhonov Regularization in General Form



$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|x\|^2$$

Prior information: The 2-norm of x is small.



$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|^2$$

Prior information: x is known to be smooth.



$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|x - x_*\|^2$$

Prior information: x is close to a known x_* .



$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|L(x - x_*)\|^2$$

Prior information: x has similar smoothing structures as x_* .

Moving Away From the 2-Norm §8.6

Tikhonov is based on penalizing the 2-norm of the solution:

$$\min_x \{ \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2 \}.$$

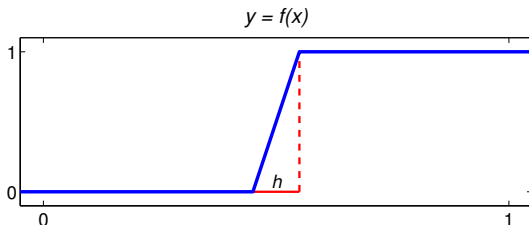
The same is true for TSVD, which can also be formulated as

$$\min \|x\|_2^2 \quad \text{subject to} \quad \|A_k x - b\|_2^2 = \min, \quad A_k = \sum_{i=1}^k u_i \sigma_i v_i^T.$$

It is the 2-norm penalization, which causes a bad reconstruction of the edges (discontinuities).

So which norm should we use to preserve the edges?

An Example Using a Continuous Function



Consider the piecewise linear function

$$f(t) = \begin{cases} 0, & 0 \leq t < \frac{1}{2}(1-h) \\ \frac{t}{h} - \frac{1-h}{2h}, & \frac{1}{2}(1-h) \leq t \leq \frac{1}{2}(1+h) \\ 1, & \frac{1}{2}(1+h) < t \leq 1 \end{cases}$$

which increases linearly from 0 to 1 in $[\frac{1}{2}(1-h), \frac{1}{2}(1+h)]$.

Norms of the First Derivative

It is easy to show that the 1- and 2-norms of $f'(t)$ satisfy

$$\|f'\|_1 = \int_0^1 |f'(t)| dt = \int_0^h \frac{1}{h} dt = 1,$$

$$\|f'\|_2^2 = \int_0^1 f'(t)^2 dt = \int_0^h \frac{1}{h^2} dt = \frac{1}{h}.$$

Note that $\|f'\|_1$ is independent of the slope of the middle part of $f(t)$, while $\|f'\|_2$ penalizes steep gradients (when h is small).

- The 2-norm of $f'(t)$ will not allow any steep gradients and therefore it produces a smooth solution.
- The 1-norm, on the other hand, allows some steep gradients – but not too many – and it is therefore able to produce a less smooth solution, and even a discontinuous solution.

Total Variation (TV) Regularization

The example motivates us to replace Tikhonov's 2-norm with the 1-norm of the first derivative, which is known as the *total variation*.

In the discrete setting:

$$\min_x \{ \|Ax - b\|_2^2 + \lambda \|Lx\|_1 \},$$

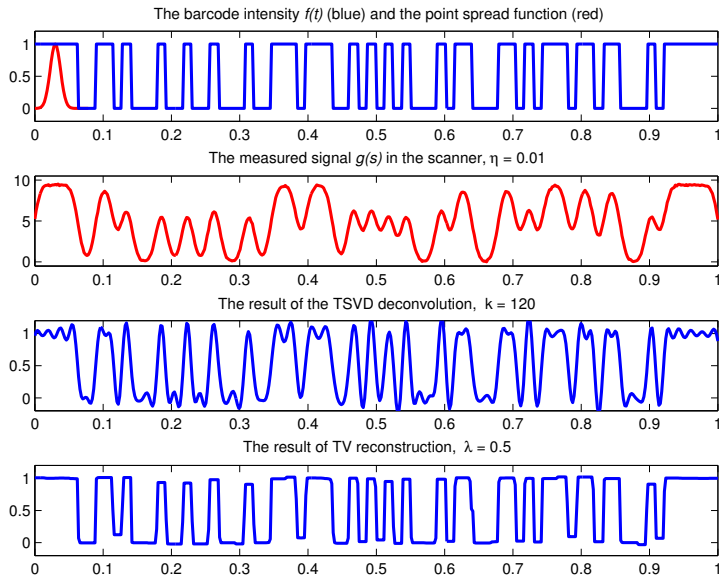
where

$$L = \begin{pmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} \in \mathbb{R}^{(n-1) \times n}$$

such that $\|Lx\|_1$ approximates the total variation $\|f'\|_1$.

The figure on the next page shows a good TV reconstruction to the barcode problem.

An Example in 1D



In two dimensions, given a function $f(\mathbf{t})$ with $\mathbf{t} = (t_1, t_2)$, we use the gradient magnitude defined as

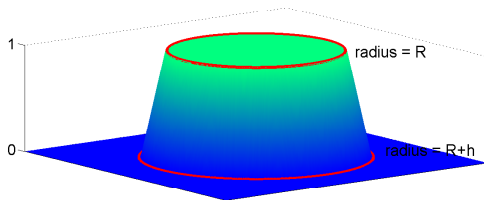
$$|\nabla f(t)| = \left(\left(\frac{\partial f(t)}{\partial t_1} \right)^2 + \left(\frac{\partial f(t)}{\partial t_2} \right)^2 \right)^{\frac{1}{2}},$$

to obtain the 2D version of the total variation $\|\nabla f(t)\|_1$. The relevant norms of $f(\mathbf{t})$ are now

$$\|\nabla f\|_1 = \int_0^1 \int_0^1 |\nabla f| dt_1 dt_2 = \int_0^1 \int_0^1 \left(\left(\frac{\partial f}{\partial t_1} \right)^2 + \left(\frac{\partial f}{\partial t_2} \right)^2 \right)^{\frac{1}{2}} dt_1 dt_2,$$

$$\|\nabla f\|_2^2 = \int_0^1 \int_0^1 |\nabla f|^2 dt_1 dt_2 = \int_0^1 \int_0^1 \left(\left(\frac{\partial f}{\partial t_1} \right)^2 + \left(\frac{\partial f}{\partial t_2} \right)^2 \right) dt_1 dt_2.$$

An Example in 2D



To illustrate the difference between these two norms, consider a function $f(\mathbf{t})$ with the polar representation

$$f(r, \theta) = \begin{cases} 1, & 0 \leq r < R \\ 1 + \frac{R}{h} - \frac{r}{h}, & R \leq r \leq R + h \\ 0, & R + h < r. \end{cases}$$

2D Example Continued

The function f is 1 inside the disk with radius $r = R$, zero outside the disk with radius $r = R + h$, and it has a linear radial slope between 0 and 1. In the area between these two disks the gradient magnitude is $|\nabla f| = 1/h$, and elsewhere it is zero.

$$\|\nabla f\|_1 = \int_0^{2\pi} \int_R^{R+h} \frac{1}{h} r \, dr \, d\theta = 2\pi R + \pi h$$

$$\|\nabla f\|_2^2 = \int_0^{2\pi} \int_R^{R+h} \frac{1}{h^2} r \, dr \, d\theta = \frac{2\pi R}{h} + \pi.$$

Similar to the one-dimensional example, we see that the total variation smoothing norm prefers steep gradients, while the 2-norm penalizes steep gradients. Furthermore, as $h \rightarrow 0$ we see that $\|\nabla f\|_1$ converges to the circumference $2\pi R$.

Total Variation in Continuous Setting (1D)

The total variation of $f : [a, b] \rightarrow \mathbb{R}$ is defined as

$$\text{TV}(f) = \int_a^b |Df| = \sup_{p \in \mathcal{P}} \sum_{i=1}^{n_p} |f(t_i) - f(t_{i-1})|,$$

where $\mathcal{P} = \{p = \{t_0, \dots, t_{n_p}\} | p \text{ is a partition of } [a, b]\}$. The supremum is taken over all partitions of the interval $[a, b]$.

- If f is differentiable, then $\text{TV}(f) = \int_a^b |f'(t)| dt$
- Example: If $f(t) = \begin{cases} -1, & \text{if } -1 \leq t \leq 0, \\ 1, & \text{if } 0 < t \leq 1 \end{cases}$, then $\text{TV}(f) = 2$.
- Bounded variation (BV) space is defined as

$$f \in \text{BV}([a, b]) \iff \text{TV}(f) < +\infty.$$

- In discrete case, $\text{TV}(f) = \sum_{i=1}^n |f'(t_i)|$.

Total variation in Continuous Setting (ND)

The total variation of $f : [a, b]^N \rightarrow \mathbb{R}$ with $\Omega = [a, b]^N$ is defined as

$$\text{TV}(f) = \sup \left\{ \int_{\Omega} f \operatorname{div} \vec{v} \, dt : \vec{v} \in C_0^1(\Omega, \mathbb{R}^N), \|\vec{v}(t)\|_2 \leq 1 \text{ for } \forall t \in \Omega \right\}$$

where $\vec{v} = (v_1, \dots, v_N)^\top$, $\operatorname{div} \vec{v} = \sum_{i=1}^N \frac{\partial v_i}{\partial t_i}$, and $C_0^1(\Omega, \mathbb{R}^N)$ is the space of continuously differentiable functions with compact support in Ω .

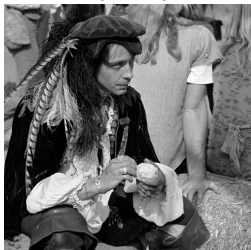
- If f is differentiable, then $\text{TV}(u) = \int_{\Omega} |\nabla f(x)|_2 \, dx$.
- In discrete case, $\text{TV}(f) = \sum_{i=1}^n |\nabla f(t_i)|_2$.
- Example: Let $\Omega = \mathbb{R}^2$ and $f = \chi_{aB^2}$ be an indicator function on the disk centered at the origin with radius $a > 0$, then $\text{TV}(f) = 2\pi a$.
- **BV space:** $BV(\Omega) = \{f \in L^1(\Omega) : \text{TV}(f) < +\infty\}$. It endowed with the norm $\|f\|_{BV(\Omega)} = \|f\|_{L^1(\Omega)} + \text{TV}(f)$ is a Banach space.
- $W^{1,1}(\Omega) \subset BV(\Omega) \subset L^1(\Omega)$

Total Variation Image Deblurring Example

The variational model with TV regularizer is

$$\min_{f \in BV(\Omega)} \|Kf - g\|^2 + \lambda TV(f)$$

Original image



Noisy and blurred image



TV deblurred image



- TV is convex.
- TV leads to stair-casing artifacts.
- TV is rotation invariant.